

Arithmetic and complexity

Is the matrix multiplication exponent two?

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because closing the exponent gap is a longstanding central barrier in algebraic algorithms; broad importance follows from matrix multiplication's role throughout NLA and computational complexity.

Topic: arithmetic complexity of dense matrix multiplication

Problem statement

Over the field $\mathbb{C}$, let $M(n)$ be the minimum number of scalar additions, subtractions, multiplications, and divisions in an arithmetic straight-line program computing every entry of AB for arbitrary $A, B \in \mathbb{C}^{n \times n}$. Programs must be defined on their intended inputs. Set $\omega = \inf\{\tau : M(n) = O(n^\tau)\}$. Is $\omega = 2$? Equivalently, for every $\varepsilon > 0$, can these products be computed in $O(n^{2+\varepsilon})$ arithmetic operations? This asks about asymptotic arithmetic cost; it does not assert an $O(n^2)$ algorithm or a floating-point stability guarantee.

Why it matters

Matrix multiplication is a basic cost driver for dense NLA.

References and status

M. Bläser, *Fast Matrix Multiplication*, Graduate Surveys 5 (2013), §§1, 5, provides the computational model. E. Dupont et al., *Improving the matrix multiplication exponent with modern optimization and AlphaEvolue* (2026), abstract and §1, reported the earlier bound $\omega < 2.371177$. OpenAI, *An Upper Bound of 9/4 for the Matrix Multiplication Exponent* (2 October 2026), reports $\omega \leq 9/4 = 2.25$ over $\mathbb{C}$. The bound is strictly above two. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked Dupont et al., August 2026, and searched for exponent-two proofs and newer matrix-multiplication bounds. Its reported bound is still strictly above two, at 2.371177. No proof of exponent two or a strict lower bound above two was located. Faster finite-size identities and improved numerical optimizations of existing bounds do not decide the asymptotic equality.

Status check — 2026-10-06

OpenAI's mathematics release points to its manuscript catalogue, which lists the 9/4 square-matrix bound over $\mathbb{C}$ and the preprint. This is a reported $O_\varepsilon(n^{9/4+\varepsilon})$ arithmetic-operation upper bound, improving the previously cited 2.371177 bound. The new claim has not been independently reviewed for this entry. Since $9/4 > 2$, it does not answer whether $\omega = 2$; **AC-01 remains open.**

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Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Verified affiliation and submission record.](#)

Sections 4–8 prove the auxiliary bound $\tilde{R}(\text{cw}_2) < 3.876919161$, also relevant to AC-04. The Section 10 scalar-tree obstruction is not a tensor-rank lower bound. Neither result proves $\omega = 2$.

Status: Open. [Report](#) · [Independent AI-agent audit and limitations](#). AI assistance disclosed; no external human review, formal verification or Lean checks. Original target and prior-source credit retained.